

Thomas Williams

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EDUCATION

University of Alabama at Birmingham

Bachelor of Arts: Computer Science

Minors: Business Administration, Information Systems

GPA: 3.8 (Magna Cum Laude)

TECHNICAL PROJECTS

A Rolling Start | Lead Gameplay Programmer and Researcher | *Godot, GDScript* |

- Developed a **3D** based mobility game in **Godot** designed to simulate powered wheelchair joystick learning for children through a gamified experience.
- Designed core game systems such as player saves and local user accounts using the **Singleton design pattern**.
- Engineered a **logging system** to capture granular performance metrics for each player including **spatial movement** and collision frequency for quantitative analysis.
- Created **unique levels** with **dynamic designs** to keep players engaged in exploration as they search through the world.
- Integrated a specialized **HID (Human Interface Device)** to simulate the feeling of a real powered wheelchair.

Real News | AI Integration and Backend Developer | *Typescript, JavaScript, AI, HTML, CSS, PostgreSQL* |

- Worked in a team to create an **AI powered** news summary aggregator that transforms the way news is consumed by using **dynamic clustering** to group related articles and AI agents to create unique briefs.
- Engineered an asynchronous **multi-agent pipeline** to orchestrate data flow and decision making for final content generation.
- Managed **end-to-end data lifecycle** to ensure integrity and consistency from ingestion to delivery.
- Implemented a **RAG (Retrieval-Augmented Generation) pipeline** to ground AI responses in verified article context **reducing hallucinations**.
- Built a **server-side AI usage system** that manages tiered quotas for anonymous and registered users via **IP hashes** and **local storage identity mapping**.

AI Powered Desktop Voice Assistant | Lead Developer | *Python, AI* |

- Developed a **Python-based** voice assistant utilizing **Mistral AI** for **natural language processing (NLP)** and command execution.
 - Automated browser actions using the **Selenium** library, enabling **voice commands** to perform searches, navigate webpages, and manage browser tabs.
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RELEVANT EXPERIENCE

Waverly Advisors | IT Intern |

April 2025 – Present

- Troubleshoot and resolve multiple complex issues including hardware, software, and network issues to maintain uptime for office operations.
- Manage the manual extraction of new client data, reformatting the structure to match company standards before moving them to the network drive for final migration.
- Mapped out detailed network topologies of 10+ offices using Visio so my team could get a better understanding of each branch's network configuration, reducing network troubleshooting time.